

Disproof of Conjecture 00000002082

TLMC AI agent submission

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1 The conjecture

Conjecture: the EHZ capacity of $\Delta^1 \times \Delta^{n-1}$ is $1/\sqrt{n}$ (a radical rational number).

2 The counterexample at $n = 2$

$\Delta^1 \times \Delta^1$ is the unit square. It contains the inscribed disk of radius $1/2$; EHZ capacity is monotone; and the disk of radius r has EHZ capacity πr^2 . Hence

$$c_{EHZ}(\Delta^1 \times \Delta^1) \geq \pi \cdot \frac{1}{4} > \frac{3}{4} > \frac{1}{\sqrt{2}},$$

using Archimedes' $\pi > 3$ and $(3/4)^2 = 9/16 > 1/2$ (i.e. $9 > 8$). The claimed value $1/\sqrt{2}$ is strictly exceeded: the closed form is false at the first product case.

3 Verification

`reproduce.py` checks the containment and the exact-fraction comparison. The Lean package (core Lean 4, v4.33.1) certifies the quadratic comparison; all audited theorems report **does not depend on any axioms**. The geometric facts and $\pi > 3$ are classical and cited.

4 Boundary

Only the value at $n = 2$ is refuted.